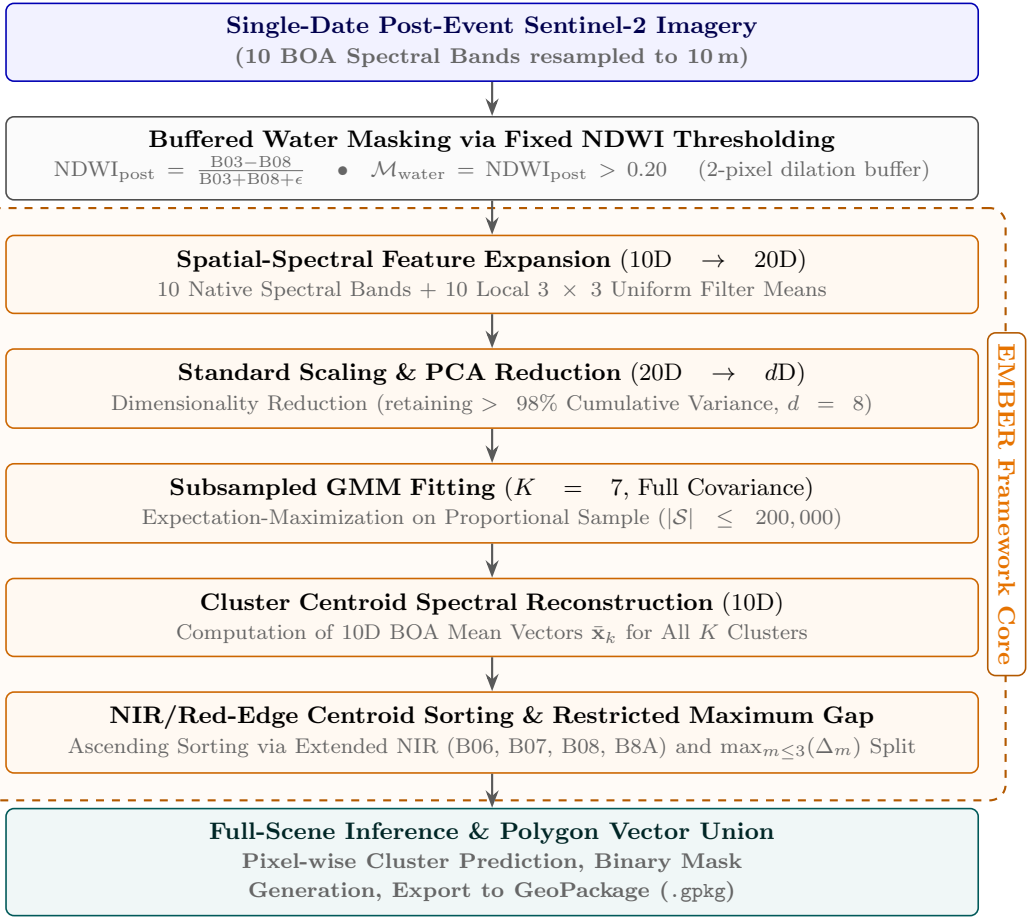

Single-Date Post-Event Sentinel-2 Imagery
(10 BOA Spectral Bands resampled to 10 m)
Buffered Water Masking via Fixed NDWI Thresholding
$\mathrm{NDWI}_{\mathrm{post}} = \frac{\mathrm{B03}-\mathrm{B08}}{\mathrm{B03}+\mathrm{B08}+\epsilon}$ • $\mathcal{M}_{\mathrm{water}} = \mathrm{NDWI}_{\mathrm{post}} > 0.20$ (2-pixel dilation buffer)
Spatial-Spectral Feature Expansion (10D → 20D)
10 Native Spectral Bands + 10 Local 3 × 3 Uniform Filter Means
Standard Scaling & PCA Reduction (20D → $d$D)
Dimensionality Reduction (retaining > 98% Cumulative Variance, $d = 8$)
Subsampled GMM Fitting ($K = 7$, Full Covariance)
Expectation-Maximization on Proportional Sample ($|\mathcal{S}| \leq 200,000$)
Cluster Centroid Spectral Reconstruction (10D)
Computation of 10D BOA Mean Vectors $\bar{\mathbf{x}}_k$ for All $K$ Clusters
NIR/Red-Edge Centroid Sorting & Restricted Maximum Gap
Ascending Sorting via Extended NIR (B06, B07, B08, B8A) and $\max_{m \leq 3}(\Delta_m)$ Split
EMBER Framework Core
Full-Scene Inference & Polygon Vector Union
Pixel-wise Cluster Prediction, Binary Mask
Generation, Export to GeoPackage (.gpkg)